

TAMIL NADU STATE BOARD - STANDARD 10 MATHEMATICS

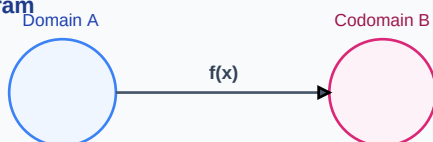
CHAPTER 5: COORDINATE GEOMETRY

PREMIUM ACADEMIC STUDY SUITE & EXAM GUIDE

Board:	Tamil Nadu State Board (TNSB)
Syllabus:	Samacheer Kalvi Textbook Curriculum aligned
Standard:	Class 10 English Medium
Subject:	Mathematics
Chapter Name:	Coordinate Geometry

Learning Objectives:

- Comprehend core theoretical concepts and exact textbook terminology.
- Apply relevant formulas and proofs to solve high-scoring board problems.
- Interpret and sketch vector diagrams or graphical models representing equations.
- Build examination readiness for 1-mark, 2-mark, and 5-mark question sets.

Function Mapping Diagram

Core Concepts & Definitions

Textbook definitions are essential for scoring fully in the short-answer section of board assessments. Review the primary definitions below:

Term	Official Definition & Context
Cartesian Coordinates	A coordinate system specifying each point uniquely in a plane by a set of numerical coordinates (x, y) .
Collinear Points	Three or more points that lie on the same straight line, yielding a triangle area of zero.
Slope of Line	A measure of the steepness of a line relative to the horizontal axis, calculated as rise over run.
Centroid	The point of intersection of the three medians of a triangle, dividing each median in a 2:1 ratio.

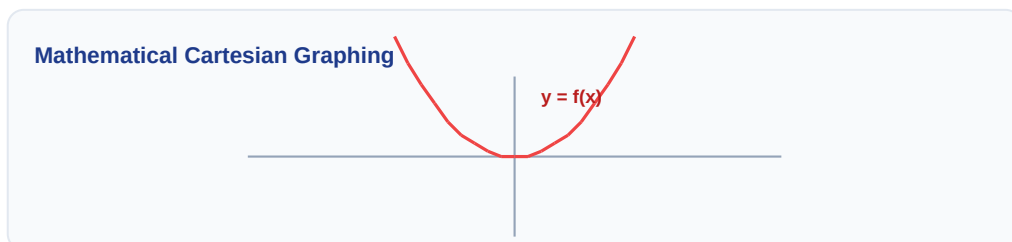
Syllabus Exam Tip: Memorize precise keywords. Examiners award maximum marks when textbook terms and correct symbols are correctly referenced.

Venn Diagram: $A \cap B$



Detailed Explanation & Formulas

Here, we analyze the core laws and formulas. Examine the conceptual illustration below to visual the relation between variables:



Essential Formulas, Laws & Identities:

- Distance Formula: $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
- Area of Triangle: $A = 0.5 * |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- Slope of a Line (m): $m = \tan(\theta)$ or $m = (y_2 - y_1)/(x_2 - x_1)$
- Slope-Intercept form: $y = mx + c$ | Point-Slope form: $y - y_1 = m(x - x_1)$
- Condition for Parallel lines: $m_1 = m_2$ | Perpendicular lines: $m_1 * m_2 = -1$

Make sure to check the dimensional consistency of all physical quantities and verify that units are correctly stated in final solutions.

Worked Examples

Review these standard board examination-style worked examples to learn proper step layout structures:

Example 1: Find the slope of a line passing through points A(5, 2) and B(3, 6).

Step-by-step Solution:

Apply slope formula: $m = (y_2 - y_1) / (x_2 - x_1)$.

Here, $m = (6 - 2) / (3 - 5) = 4 / (-2) = -2$.

The negative sign indicates the line slopes downwards to the right.

Linear Algebraic Matrix Representation

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix}$$

HOTS (Higher Order Thinking Skills) Challenge

Challenge yourself with these complex, multi-concept problems designed to test deeper conceptual understanding and mathematical reasoning:

HOTS Challenge 1: Find the equation of a line passing through (2, 5) and perpendicular to the line joining (0, 0) and (4, 2).

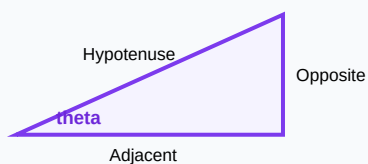
Analytical Solution Strategy:

Slope of first line: $m_1 = (2 - 0) / (4 - 0) = 2/4 = 0.5$.

Perpendicular slope: $m_2 = -1 / m_1 = -2$.

Equation using point-slope form with (2, 5): $y - 5 = -2(x - 2) \Rightarrow y - 5 = -2x + 4 \Rightarrow 2x + y - 9 = 0$.

Right-Angled Triangle Geometry



Practice Worksheet

Test your skills by attempting these sample exam problems independently. Draw diagrams where appropriate:

Q1 (1-Mark): Find the centroid of a triangle with vertices $(2, -4)$, $(-3, 6)$, and $(7, 10)$.

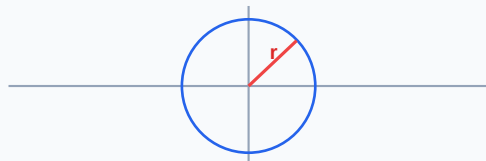
Q2 (2-Mark): Check if points $A(1, 2)$, $B(2, 3)$, and $C(3, 4)$ are collinear.

Q3 (2-Mark): Write the equation of the line passing through $(1, 1)$ with slope 3.

Q4 (5-Mark): Find the slope of a line perpendicular to $y = 4x - 10$.

Q5 (5-Mark): Evaluate the area of a triangle whose vertices are $(0,0)$, $(4,0)$, and $(0,3)$.

Circle with Radius Vector on Plane



Real-World Applications & Connections

This chapter connects directly with modern industrial engineering, computation, and natural systems in numerous ways:

Application Case: GPS Mapping Systems

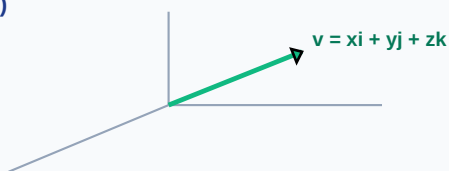
Global Positioning Systems (GPS) utilize cartesian coordinate lines to locate addresses and plan shortest path routes.

Application Case: Game Physics

Collision vectors and player motion paths are calculated continuously in 2D coordinate space using line slopes.

Cross-Curricular Link: Concepts learned in this unit are heavily applied in other subjects like Physics (equations of motion, field vectors) and Data Analytics.

3D Vector Projections (x, y, z)



Revision Summary & Strategy

Use this checklist to self-evaluate your exam preparation status before final tests:

Key Recall Tips:

- ✓ Always write down 'Given Parameters' first to ensure partial marks are locked in.
- ✓ Pay attention to signs (+ / -) when performing algebraic operations or force mappings.
- ✓ Check that you have written the correct units (e.g. Joules, Tesla, cm^3 , N) for the final numeric answer.
- ✓ Draw neat diagrams using a sharp pencil and ruler for geometry or circuit layouts.

Syllabus Checklist:

- ☐ Read and memorized core definitions on Page 2.
- ☐ Reviewed critical formulas and relations on Page 3.
- ☐ Solved worked examples on Page 4.
- ☐ Attempted worksheet problems on Page 6.

CONGRATULATIONS! YOU HAVE COMPLETED THIS SYLLABUS STUDY GUIDE!

Practice consistently, verify answers with standard textbook methods, and take the topic test in the Nexora Learning app to gauge progress.
Good luck!

Probability Tree Outcome Diagram

